DataWais MCP

Dataset validation and controlled cleaning through a secure MCP file server

Midterms PT 1 submission report
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Prepared: 17 October 2026

Repository: https://github.com/tatsu-06/DataWais-MCP

Project overview

DataWais is a working Model Context Protocol file-server connector for small CSV datasets. A connected AI assistant can inspect files, apply declared validation rules, locate duplicates and create an audited cleaned copy. Project state is saved in files so a fresh agent session can continue the task.

The recorded demonstration completed a multi-step task across two fresh AI agent contexts. It retained 6 of 8 input records, recorded 2 exclusions, applied one explicit label correction and preserved the original file. The implementation passes 50 automated tests and 18 public MCP protocol and security checks.

Problem and intended use

Missing fields, duplicate examples and inconsistent labels can disrupt dataset preparation. DataWais gives AI engineering students and developers repeatable checks and a documented handoff instead of relying on details held only in a chat. The sample university-question dataset is synthetic and contains no student records.

Alignment with the assignment

Requirement	Implementation
Sandboxed file access	One workspace root; path escapes and links rejected.
At least eight tools	Ten tools including list, read, write, str_replace, delete and search.
HTTPS and process management	Authenticated Cloudflare tunnel and pm2 supervision.
Persistent handoff	PROJECT_STATE, PLAN_LOG, CHECKPOINTS and DECISIONS.
Real client and two sessions	Official SDK client, MCP Inspector and native Codex verification.
Documentation and evidence	README, technical report, tests and full tool-call transcripts.

Course reference: MCP Week 6 Presentation 2, printed slides 16 to 20. This report documents the completed build and recorded run. The assessed five-minute classroom presentation is a separate delivery step.

Architecture and tool design

The AI host handles the user conversation and chooses actions. Its MCP client initializes a connection, discovers tool schemas and sends calls. DataWais executes those calls against a restricted workspace and returns structured results or an error result.

Layer	Choice and purpose
Runtime and SDK	Python with the official MCP Python SDK 1.30.0, pinned for reproducibility.
Validation and data	Pydantic argument models; standard Python CSV parsing for quoted commas and multiline values.
Transport and serving	MCP Streamable HTTP served locally by Uvicorn; HTTPS supplied by Cloudflare Tunnel.
Supervision and history	pm2 manages the server and tunnel; Git records source and documentation changes.
Verification	pytest, a real SDK-based MCP client, official MCP Inspector and native Codex tools.

Ten single purpose tools

Tool	Purpose
list_files	Paginate one permitted directory.
read_file	Read a bounded UTF-8 slice and its SHA-256.
write_file	Create or hash-guard an editable report or handoff file.
str_replace	Replace exactly one unique match in an editable text file.
delete_file	Admin-only, confirmed and hash-guarded deletion of an eligible generated file.
search_files	Search literal text and return bounded excerpts.
inspect_dataset	Return columns, counts, missing values, samples and source hash.
validate_dataset	Apply an operator-supplied rules file and report issues.
find_duplicates	Identify exact duplicate records using selected columns.
create_clean_copy	Create a new cleaned CSV, excluded-record file and audit report atomically.

The inspector receives 7 tools, the cleaner 9 and the admin all 10. Role scope is enforced by separate server tool registries, not only by instructions. A read-only project-state resource and a reusable continuation prompt supplement the tools.

Security and implementation decisions

Workspace boundaries

Every requested path is checked against one configured root. Absolute paths, parent traversal, alternate streams, device names, hidden paths, symbolic links, junctions and Windows reparse points are rejected. Agents cannot access the user profile or the private runtime folder through these tools.

Raw datasets and operator rules are read-only to all agent roles. Generic writes are limited to approved text files in reports, handoff and notes folders. An inspector cannot clean data or delete files. The cleaner cannot delete files.

Authentication and change control

Remote calls require a random role-specific bearer token over HTTPS. The managed server receives the three role tokens through environment variables. Local provisioning and client authentication use ignored private runtime files. Tokens are excluded from the repository, shared archive and server audit contents.

Overwriting an editable file requires its current SHA-256. Exact replacement must match once. Deletion requires the admin role, a current hash and explicit confirmation. Cleaning requires current hashes for the source, validation rules and cleaning policy; duplicate output names are rejected.

Deterministic cleaning

The cleaning policy trims selected fields, applies explicit label mappings, excludes invalid records and keeps the first remaining duplicate-key occurrence. It never invents missing values or guesses label corrections. Each run creates cleaned.csv, excluded.json and report.json in a new run directory; excluded originals and reasons remain reviewable.

Bounded inputs and useful failures

The first version supports UTF-8 CSV files up to 5 MiB, 10,000 records and 100 columns. Reads, search results and samples are bounded. Strict schemas reject unknown arguments. Tool errors use isError results with recovery guidance rather than exposing raw stack traces.

Tradeoffs and limitations

The system assumes a trusted local filesystem owner and one managed writer process. Its lock does not provide cross-process isolation or defeat a hostile local user racing filesystem changes. Prototype token authentication does not provide full OAuth or per-user revocation. Validation establishes compliance with declared rules, not factual accuracy, fairness or training suitability.

The quick-tunnel URL can change after a restart. A wrapper updates the exact allowed hostname and private client settings automatically. Remote access requires the host computer to remain awake and online; a permanent URL would require an operator-owned named tunnel or domain.

Recorded two session demonstration

Task and shared memory

The task prepared a question-classification CSV with allowed labels enrollment, schedule and payment. The two agents communicated through PROJECT_STATE, PLAN_LOG, CHECKPOINTS and DECISIONS. The continuation agent received the project location and objective, then read the saved handoff; the earlier conversation and findings were not copied into its context.

Stage	Actual operations and result
Session one inspector	Discovered 7 available tools; read shared state and rules; inspected and validated 8 records; saved findings, input hashes and the continuation plan. It stopped before cleaning.
Session two cleaner	Discovered 9 tools; read the handoff; verified all three input hashes; created one cleaned run; independently validated the output and checked the unchanged source.
Persistent outcome	Saved the final report and refreshed all four shared handoff files using current hash guards. The source and completed outputs remained available after server recovery.

Measured dataset results

Measure	Recorded result
Input records	8
Retained records	6
Excluded records	2: one duplicate and one missing question
Explicit corrections	One paymnt to payment mapping and one whitespace correction
Final invalid records and duplicates	0 invalid records; 0 duplicate records
CSV edge cases	Quoted comma and embedded newline preserved

Original SHA-256

`9494e4d2774094d4f3480c958a308487ed62e2ba941b5dff1cd9629248febef7`

Cleaned SHA-256

`c26b3020b271c1587967e24499436990cc076d655f669b95cdb65ad98a6191de`

Observed efficiency and error recovery

The inspector used 20 tool calls with no tool errors. The cleaner used 28 calls and handled one missing-directory result before the first output run was created. Both transcripts retain that evidence. The initializer now creates empty output directories for later runs. A timed live rehearsal should reduce redundant reads and batch independent operations where supported.

Full, unedited inputs and results: Session one transcript and Session two transcript.

Verification setup and live demonstration

Verification	Evidence and outcome
Automated suite	50 tests passed. Covers CSV handling, path protections, version guards, rule provenance, credential configuration and tunnel-host updates.
Public MCP checks	18 checks passed through normal DNS with TLS certificate verification. Includes authentication, role scope, strict schemas, file guards, resource and prompt access.
MCP Inspector	Admin tool discovery exposed all 10 tools; an actual dataset inspection succeeded.
Native Codex connection	Read shared state, inspected the source, validated the cleaned copy, searched records and refreshed deployment state.
Recovery	pm2 recovered a stopped server; the saved handoff persisted. A current-user sign-in task starts the configured installation.

Reproduce from the repository

Install Python 3.11 or newer, Node.js and Git. Clone the repository and open its directory. The README contains the complete installation and HTTPS setup instructions.

```
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -e ".[test]"
.\.venv\Scripts\python.exe scripts\init_demo.py
.\.venv\Scripts\python.exe -m datawais.server
```

The MCP endpoint is http://127.0.0.1:8765/mcp. Choose the inspector or cleaner role token from the locally generated private credentials. Use MCP Inspector or the included SDK client. The public URL is generated by the tunnel and stored privately; credentials are never included in this document.

Five minute live demonstration plan

Time	What to demonstrate
0:00 to 0:40	State the problem and show real MCP discovery.
0:40 to 2:00	Inspect and validate the CSV in session one; save a handoff.
2:00 to 2:30	Switch to a fresh session and load shared state.
2:30 to 4:15	Create the cleaned copy; verify output and unchanged source.
4:15 to 5:00	Show a rejected path escape, scoped tools and audit evidence.

Reference material

Project README; Technical report; Official MCP Python SDK; Official MCP Inspector.

Appendix session 1 tool call record

Session identifier: agent-1-inspection
Role: inspector | Tool calls: 20 | Tool errors: 0

Each row corresponds to an actual tool call in chronological order. Input targets and results are condensed for readability; full arguments and returned content remain in the linked JSONL transcript. Tool discovery was recorded separately.

#	Tool	Input target	Observed result
1	read_file	handoff/PROJECT_STATE.md	Read content and current hash.
2	read_file	handoff/CHECKPOINTS.md	Read content and current hash.
3	read_file	handoff/DECISIONS.md	Read content and current hash.
4	inspect_dataset	raw/university_questions.csv	8 records inspected.
5	find_duplicates	raw/university_questions.csv	1 duplicate records.
6	read_file	rules/questions.json	Read content and current hash.
7	read_file	handoff/PLAN_LOG.md	Read content and current hash.
8	validate_dataset	raw/university_questions.csv	valid=False; invalid=2; duplicates=1
9	read_file	rules/cleaning_policy.json	Read content and current hash.
10	read_file	raw/university_questions.csv	Read content and current hash.
11	list_files	.	Listed 4 entries.
12	list_files	handoff	Listed 4 entries.
13	write_file	handoff/INSPECTION_REPORT.md	Saved with hash guard when replacing.
14	write_file	handoff/INSPECTION_REPORT.md	Saved with hash guard when replacing.
15	write_file	handoff/DECISIONS.md	Saved with hash guard when replacing.
16	write_file	handoff/CHECKPOINTS.md	Saved with hash guard when replacing.
17	write_file	handoff/PLAN_LOG.md	Saved with hash guard when replacing.
18	write_file	handoff/PROJECT_STATE.md	Saved with hash guard when replacing.
19	read_file	handoff/PROJECT_STATE.md	Read content and current hash.
20	inspect_dataset	raw/university_questions.csv	8 records inspected.

Full transcript: agent-1-transcript.jsonl

Appendix session 2 tool call record

Session identifier: agent-2-cleaning
Role: cleaner | Tool calls: 28 | Tool errors: 1

Each row corresponds to an actual tool call in chronological order. Input targets and results are condensed for readability; full arguments and returned content remain in the linked JSONL transcript. Tool discovery was recorded separately.

#	Tool	Input target	Observed result
1	read_file	handoff/PROJECT_STATE.md	Read content and current hash.
2	list_files	handoff	Listed 5 entries.
3	read_file	handoff/INSPECTION_REPORT.md	Read content and current hash.
4	read_file	handoff/DECISIONS.md	Read content and current hash.
5	read_file	handoff/PLAN_LOG.md	Read content and current hash.
6	read_file	handoff/CHECKPOINTS.md	Read content and current hash.
7	read_file	rules/questions.json	Read content and current hash.
8	read_file	rules/cleaning_policy.json	Read content and current hash.
9	inspect_dataset	raw/university_questions.csv	8 records inspected.
10	list_files	datasets/cleaned	Error: directory did not exist; recovered before cleaning.
11	list_files	datasets	Listed 1 entries.
12	create_clean_copy	pt1-demo-1	Created run: 8 input, 6 retained, 2 excluded.
13	validate_dataset	cleaned/pt1-demo-1/cleaned.csv	valid=True; invalid=0; duplicates=0
14	find_duplicates	cleaned/pt1-demo-1/cleaned.csv	0 duplicate records.
15	inspect_dataset	cleaned/pt1-demo-1/cleaned.csv	6 records inspected.
16	read_file	cleaned/pt1-demo-1/cleaned.csv	Read content and current hash.
17	read_file	cleaned/pt1-demo-1/excluded.json	Read content and current hash.
18	read_file	cleaned/pt1-demo-1/report.json	Read content and current hash.
19	inspect_dataset	raw/university_questions.csv	8 records inspected.
20	write_file	handoff/CLEANING_REPORT.md	Saved with hash guard when replacing.
21	read_file	handoff/PROJECT_STATE.md	Read content and current hash.
22	write_file	handoff/PROJECT_STATE.md	Saved with hash guard when replacing.
23	read_file	handoff/PLAN_LOG.md	Read content and current hash.
24	write_file	handoff/PLAN_LOG.md	Saved with hash guard when replacing.
25	read_file	handoff/DECISIONS.md	Read content and current hash.
26	write_file	handoff/DECISIONS.md	Saved with hash guard when replacing.
27	read_file	handoff/CHECKPOINTS.md	Read content and current hash.
28	write_file	handoff/CHECKPOINTS.md	Saved with hash guard when replacing.

Full transcript: agent-2-transcript.jsonl

The recorded task was executed by fresh AI agent contexts selecting MCP operations through the official SDK client. The separate scripted HTTP checks are labeled as verification and are not presented as autonomous agent sessions.